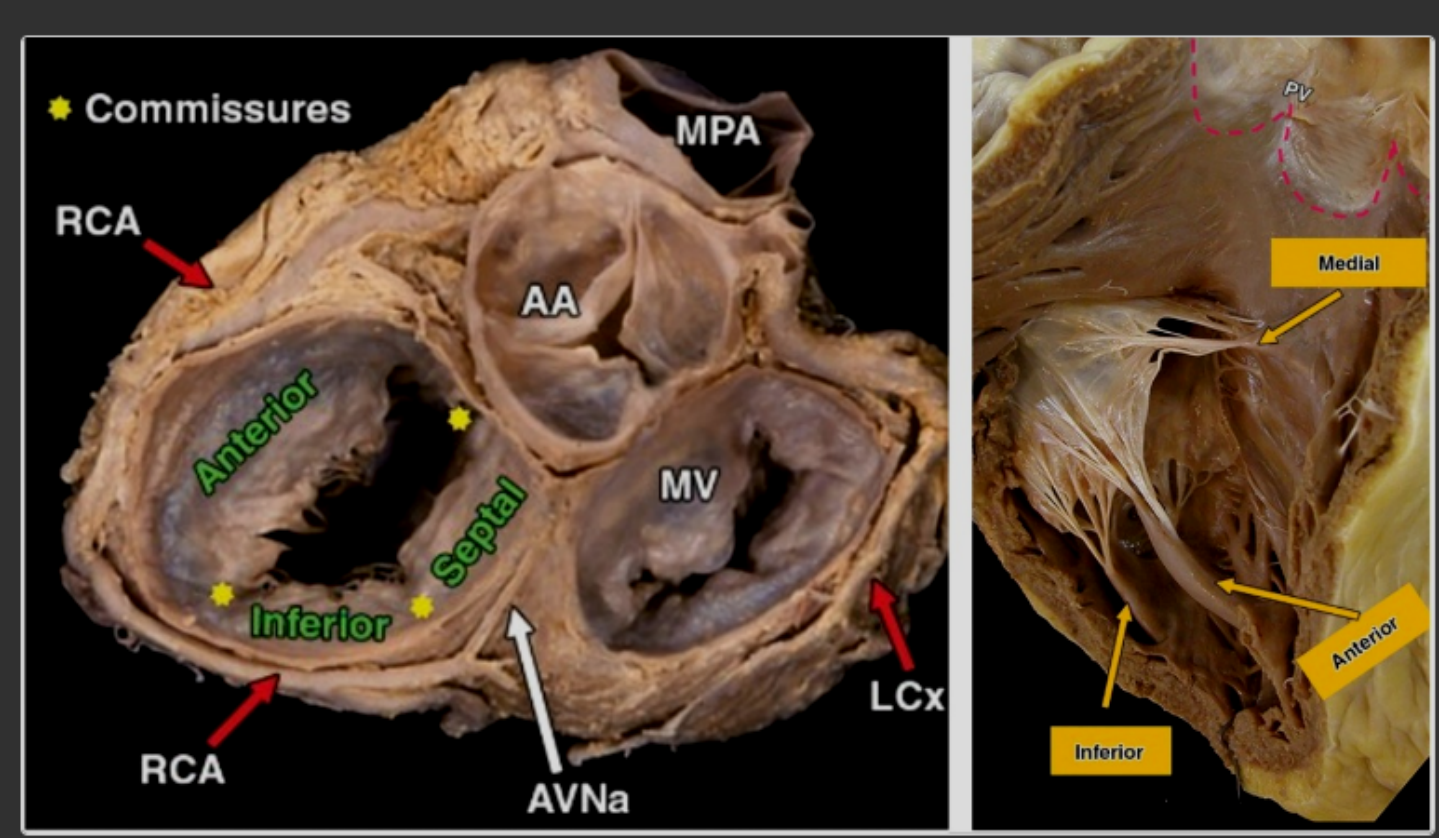

Figure 1. Healthy cadaveric Tricuspid Valve, with anterior, septal/medial, and inferior leaflets. Papillary muscles and chordae tendineae viewed inside RV.

What Is Tricuspid Regurgitation? (LO 1)

Tricuspid regurgitation is the most common right heart valve abnormality. It is retrograde flow of blood from the right ventricle (RV) through the tricuspid valve into the right atrium (RA) during systole.

Epidemiology

Symptomatic Tricuspi than symptomatic Mitral Regurgitation, population. (Mild,

What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Mitral Regurgitation and Mitral Valve Prolapse

Last updated February 20, 2026 17 min read ScholarRx

Adapted by: Diptiman Bose, Patrick Brooks, Phil Riley

Editors: Chelsey Deel, Jordan Finn, Russell Johnson

Adapted from: Mitral Regurgitation

Listen to this Brick
0:00 / 24:50 1x

Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Describe the clinical presentation of Mitral Regurgitation and Mitral Valve Prolapse.
2. Explain the etiologies and pathophysiology of acute and chronic Mitral Regurgitation and Mitral Valve Prolapse.
3. Explain how itral Valve

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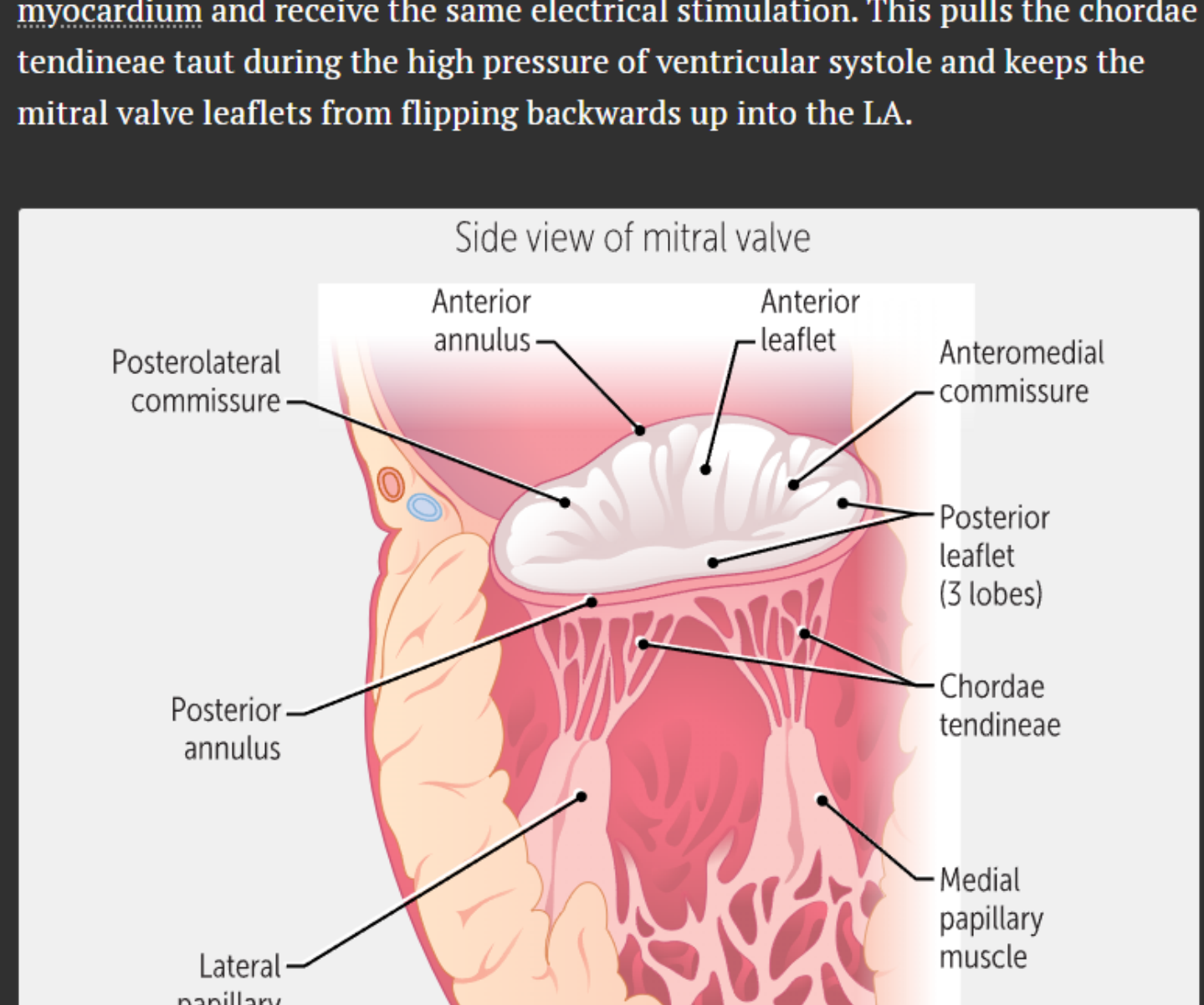
myocardium and receive the same electrical stimulation. This pulls the chordae tendineae taut during the high pressure of ventricular systole and keeps the mitral valve leaflets from flipping backwards up into the LA.



QUIZ

Figure 1. Mitral Valve anatomy

surgery can be performed, although this therapy cannot definitively treat MR. β-Blockers (B) can help reduce symptoms of heart failure but are not a definitive treatment for MR. Balloon valvuloplasty (C) is a treatment for valvular stenosis (see other lessons).

3. How is the diagnosis of mitral regurgitation confirmed?

A. ○ Auscultation of the heart
B. ○ Nuclear Medicine Cardiac stress test
C. ○ Computed Tomography
D. ● Echocardiography
E. ○ X-ray

Hide Explanation

The correct answer is echocardiography (D), which confirms and determines the severity of MR. Auscultation of the heart (A) may reveal an apical systolic murmur characteristic of MR, but an echocardiogram is still needed for confirmation. NM stress test (B), Computed Tomography (C) and X-ray (E) do not evaluate valve function.

4. Patients with seve typically present with which of the following?

A. ○ cardiac auscultation revealing a mid systolic click and early systolic murmur
B. ● slowly progressing symptoms of heart failure
C. ○ sudden, severe heart failure with no prior symptoms
D. ○ symptoms of heart failure as soon as MVP develops

Hide Explanation

The correct answer is (B). MVP, when it progresses, results in chronic MR, not acute (C). Most cases of MVP remain asymptomatic (D). The murmur of MR from MVP would occur after the prolapse/midsystolic click, as a late systolic murmur (A).

How would you rate this Brick?

References

- Douedi S, Douedi H. Mitral regurgitation. In: *StatPearls*. 27, 2022. https://www.healio.com/cardiology/learn-the-heart/cardiology-review/topic-reviews/introduction-and-etiology
- Morningstar JE, Nieman A, Wang C, Beck T, Harvey A, Norris RA. Mitral Valve Prolapse and Its Motley Crew-Syndromic Prevalence, Pathophysiology, and Progression of a Common Heart Condition. J Am Heart Assoc. 2021 Jul 6;10(13):e020919. doi: 10.1161/JAHA.121.020919. Epub 2021 Jun 22. PMID: 34155898; PMCID: PMC8403286. https://www.ahajournals.org/doi/10.1161/JAHA.121.020919

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Describe the clinical presentation of Mitral Regurgitation and Mitral Valve Prolapse.~~
02 ☑ ~~Explain the etiologies and pathophysiology of acute and chronic Mitral Regurgitation and Mitral Valve Prolapse.~~
03 ☑ ~~Explain how to diagnose Mitral Regurgitation and Mitral Valve Prolapse.~~
04 ☑ ~~Explain the management of Mitral Regurgitation and Mitral Valve Prolapse.~~

Objective Confidence: 4/4

asymptomatic, trivial TR is commonly seen on Echocardiograms.)

Causes

Tricuspid regurgitation is categorized based on the initial event. Those that cause regurgitation first are primary causes. Those that cause right ventricular dilation first and regurgitation second are secondary causes.

Primary Tricuspid Regurgitation. Diseases that directly damage the tricuspid valve will weaken the valve and prevent its proper closure, leading to primary tricuspid regurgitation:

- **Infective endocarditis (IE) in IV drug use** involves bacteria colonizing valves and forming vegetations preventing valve cusps from closing completely - resulting in regurgitation. Fever and a new heart murmur of TR in an individual with signs of IV drug use should raise suspicion for IE. IE is the most common cause of TR in individuals who use injectable drugs, and TR is the most common valve disease associated with IE from IV drug abuse.
- **Pacemaker leads** traveling from the RA to the RV pass through the orifice of the tricuspid valve and can sometimes prevent proper closure of the leaflets.
- **Ebstein anomaly** is a congenital defect that causes abnormal positioning and functioning of the tricuspid valve. Maternal use of Lithium during pregnancy increases the risk of this malformation. (Figure 2)
- **Carcinoid syndrome** is caused by neuroendocrine tumors secreting vasoactive hormones, especially serotonin. Serotonin induces fibrous tissue deposition on the right heart valves (and symptoms of diarrhea, flushing, and bro nore affected because the lung inactivates the vasoactive substances before they can

CASE CONNECTION BACK TO INTRODUCTION ↑

The team concludes that TJ has infective endocarditis of the tricuspid valve due to intravenous drug use. *S aureus* is the most common pathogen in this disease, colonizing and directly damaging the tricuspid valve. This leads to incomplete closure of the valve and regurgitant blood flow into the right atrium, superior vena cava, and inferior vena cava.

Any maneuver that increases venous return to the right heart, such as inspiration and leg raising, accentuates the murmur. IV antibiotics are started and a referral made to a drug rehabilitation program.

Summary

- Tricuspid Regurgitation is the most common right heart valve abnormality.
- Common causes include right ventricular dilation, pulmonary hypertension, and infective endocarditis in individuals who use injectable drugs.
- Tricuspid Regurgitation causes volume overload (congestion) of the right heart, which results in a dilated right atrium and ventricle, with possible atrial fibrillation.
- Mild-to-moderate Tricuspid Regurgitation is usually asymptomatic, whereas severe regurgitation presents with RHF.
- Tricuspid valve replacement is indicated for patients who do not respond to medical management alone.
- Tricuspid Stenos ular defects such as Mitral Stenosis and Aortic valve disease in the setting of rheumatic heart

What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Tricuspid Regurgitation and Tricuspid Stenosis

Last updated December 26, 2025 17 min read ScholarRx

Adapted by: Diptiman Bose, Patrick Brooks

Editors: Jordan Finn, Russell Johnson

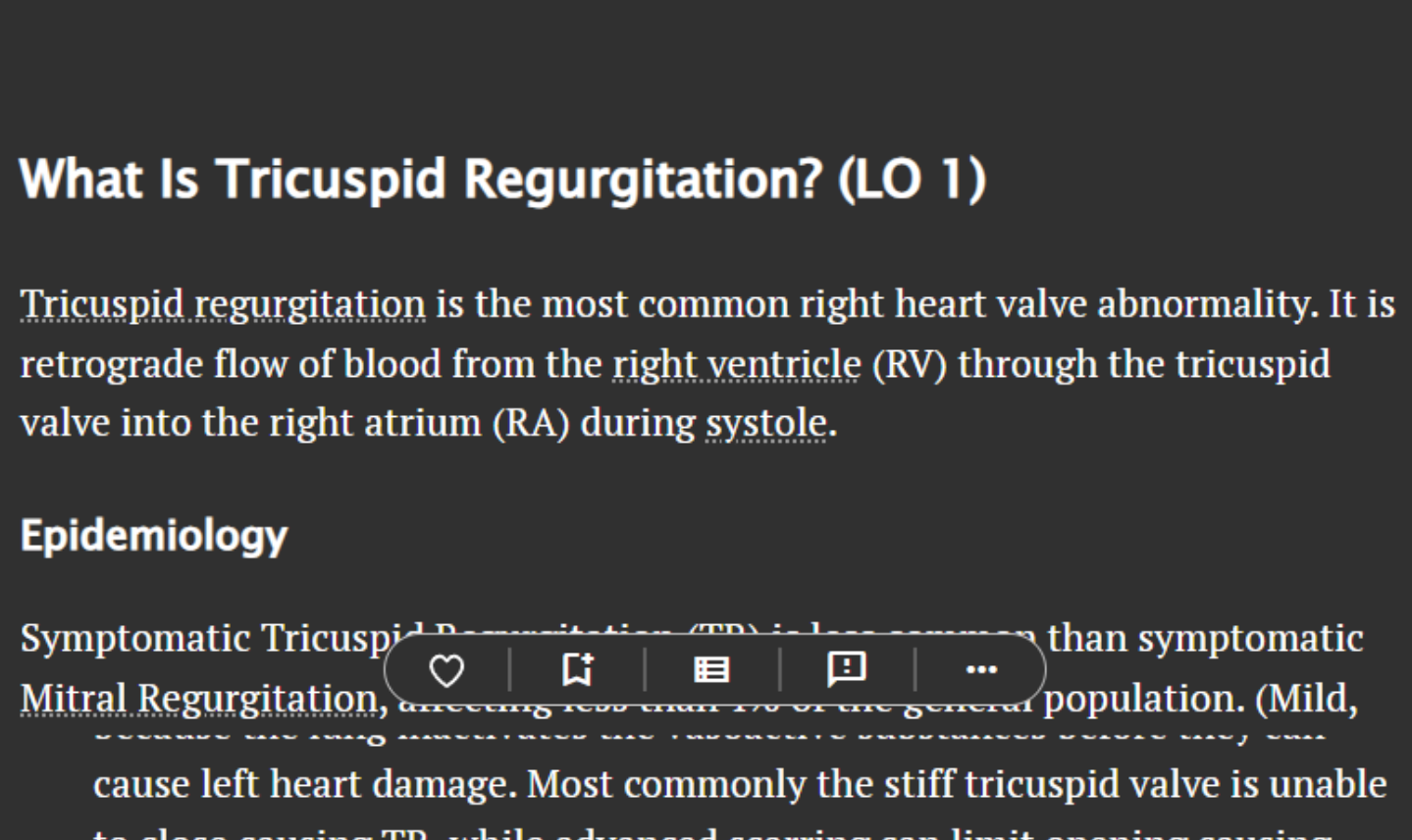
Adapted from: Tricuspid Regurgitation and Tricuspid Stenosis

Listen to this Brick
0:00 / 27:01 1x

Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Describe the etiologies and pathophysiology of tricuspid regurgitation.
2. Describe the clinical presentation, diagnosis, and management of tricuspid regurgitation.
3. Describe the uspid stenosis.



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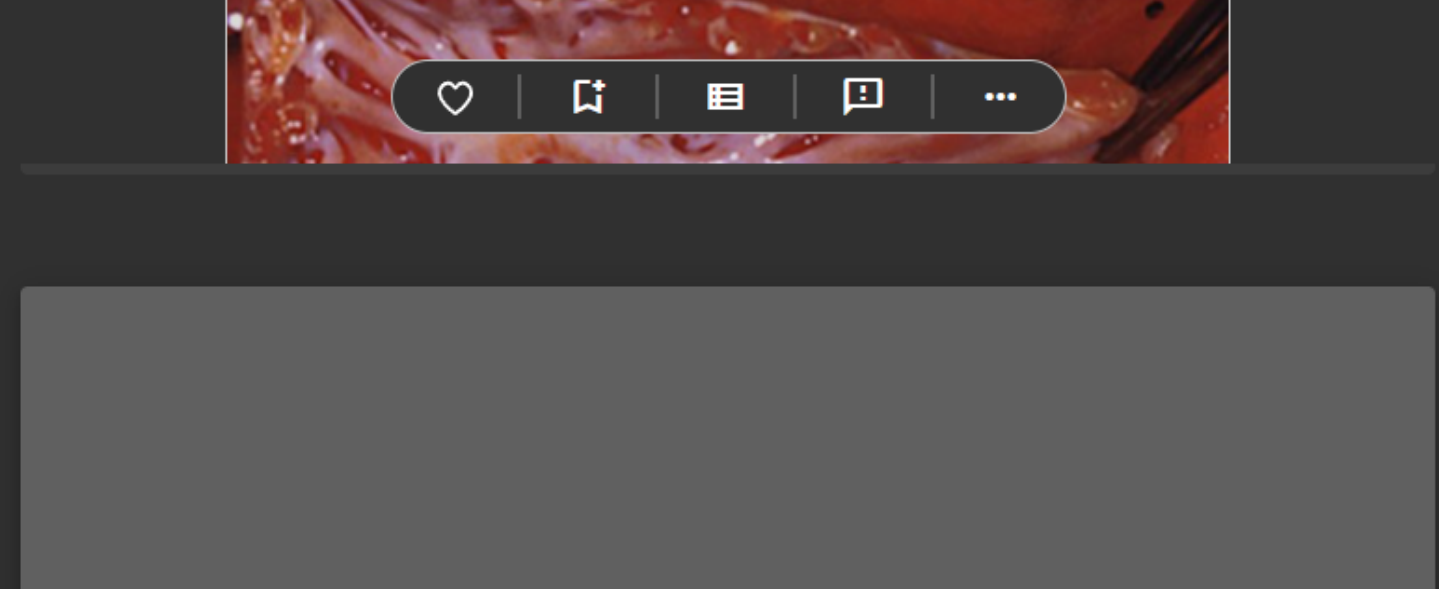
cause left heart damage. Most commonly the stiff tricuspid valve is unable to close causing TR, while advanced scarring can limit opening causing stenosis. (Figure 3)

- **Rheumatic heart disease** more rarely affects the tricuspid valve (10% of cases) than left sided heart valves. TR typically occurs in conjunction with mitral or aortic valve disease. Left sided heart valves face much higher velocity of blood flow and pressures, and this mechanical strain makes them more vulnerable to damage and subsequent scarring.
- **Connective Tissue Disorders** can cause any valvular regurgitation
- **Trauma** can potentially injure any heart valve, causing regurgitation.



Ebstein's anomaly is a congenital heart condition which results in an abnormality of the tricuspid valve. In this condition the tricuspid valve is elongated and displaced downward towards the right ventricle. The abnormality causes the tricuspid valve to leak blood backwards into the right atrium.

Figure 2. Ebstein Anomaly

For individuals who use IV drugs, which valve disorder is most commonly involved in Infective Endocarditis?

Secondary Tricuspid Regurgitation. In secondary tricuspid regurgitation, a **dilated RV** prevents a structurally normal tricuspid valve from closing. When a persistently increased RV end systolic volume leads to RV dilation, it stretches the tricuspid annulus or results in papillary muscle displacement and tethering preventing proper closure of the valve leaflets. (Figure 4) **Basic causes of RV dilation are:**

- **Pulmonary Artery Hypertension from Left Heart Failure (including left heart valvular disease) or Pulmonary disease** increases resistance to ejection of blood (afterload) from the RV, increasing its volume. Left Heart Failure is therefore the most common cause of TR.
- **RV Infarction o**